

Sprint 4

Sprint Goal

Stabalise the backend application through comprehensive integration testing while beginning frontend development and completing the core system design documentation.

ID	User Story	Priority	Story Points	Status
PB-14	Integration Testing	Must Have	5	Done
PB-15	Lifecycle workflow testing	Must Have	5	Done
PB-16	React Login and Application routing	Must Have	5	Done
PB-17	Complete system design documentation	Should Have	5	Done

Development Tasks

- Develop integration tests for container API endpoints
- Validate lifecycle expiry and supervisor reset workflows
- Complete backend integration testing
- Implement React Login Page
- Configure application routing
- Complete UML diagrams and system documentation
- Verify backend stability through automated tests

GitHub Evidence

- **19 July 2026** – Add integration tests for container API endpoints.
- **19 Jul 2026** – Add lifecycle expiry validation and supervisor reset workflow tests.
- **20 Jul 2026** – Complete backend integration test suite (28 passing tests).
- **20 Jul 2026** – Complete integration test suite.
- **20 Jul 2026** – Implement React login and application routing.
- **20 Jul 2026** – Complete system design documentation and UML diagrams.

Definition of Done

- Integration tests complete
- Backend workflow validated
- React authentication interface operational
- Application routing configured
- UML documentation completed
- Source code committed to GitHub
- Sprint deliverables integrated successfully

Sprint Review

Sprint 4 focused on improving the quality and reliability of the application through comprehensive integration testing while laying the foundations for the React frontend. Automated integration test was developed to validate API endpoints, lifecycle expiry rules and supervisor approval workflows. Also, the front-end authentication interface and application routing were implemented, and the UML diagrams and design documentation were completed.

What Went Well

- Integration testing significantly increased confidence in the application's reliability
- Backend business rules were successfully validated through automated tests
- React application structure was established
- System documentation accurately reflected the implementation solution

Challenges

- Developing comprehensive integration tests for complex workflow scenarios
- Ensuring lifecycle expiry and reset behaviour remained consistent
- Configuring frontend routing while maintaining secure authentication

Focus for Sprint 5

- Develop dashboard interface
- Implement container management pages
- Develop approval management interface
- Integrate frontend with backend APIs
- Implement operational dashboard components